

Homework II — Solutions

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Problem 1

1.A)

The definition of the canonical conjugated momenta is,

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}, \quad \pi^* = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^*}, \quad \dot{\phi} = \partial_0 \phi \quad (1.1)$$

it's straightforward to obtain then from the Lagrangian,

$$\pi = \dot{\phi}^*, \quad \pi^* = \dot{\phi} \quad (1.2)$$

The definition of the Hamiltonian density is,

$$\mathcal{H} = \pi \dot{\phi} + \pi^* \dot{\phi}^* - \mathcal{L} \quad (1.3)$$

$$\mathcal{H} = \pi \pi^* + \pi^* \pi + \partial_\mu \phi^* \partial^\mu \phi + m^2 \phi^* \phi \quad (1.4)$$

$$\mathcal{H} = 2\pi^* \pi - \dot{\phi}^* \dot{\phi} + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi \quad (1.5)$$

$$\mathcal{H} = 2\pi^* \pi - \pi \pi^* + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi \quad (1.6)$$

$$\mathcal{H} = \pi^* \pi + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi \quad (1.7)$$

the Hamiltonian is then given by the space integral,

$$H = \int d^3 \mathbf{x} \mathcal{H} \quad (1.8)$$

$$H = \int d^3 \mathbf{x} [\pi^* \pi + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi] \quad (1.9)$$

1.B)

First let's check it is a symmetry, the transformation law is,

$$\begin{cases} x & \rightarrow x' = x \\ \phi(x) & \rightarrow \phi'(x') = e^{ia} \phi(x) \\ \phi^*(x) & \rightarrow \phi^{*'}(x') = e^{-ia} \phi^*(x) \end{cases} \quad (1.10)$$

substituting back in the action,

$$S'_a[\phi'] = \int d^4 x' (-\partial'_\mu \phi^{*'} \partial^{\mu'} \phi' - m^2 \phi^{*'} \phi') \quad (1.11)$$

$$S'_a[\phi'] = \int d^4 x (-\partial_\mu e^{-ia} \phi^* \partial^\mu e^{ia} \phi - m^2 e^{-ia} \phi^* e^{ia} \phi) \quad (1.12)$$

$$S'_a[\phi'] = \int d^4 x (-\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi) = S[\phi] \quad (1.13)$$

indeed is a symmetry, then, we can use Noether's theorem,

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi_I(x)} \frac{d\phi'_I(x)}{da} \Big|_{a=0} + \mathcal{L} \frac{dx'^\mu}{da} \Big|_{a=0} \quad (1.14)$$

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{d\phi'(x)}{da} \Big|_{a=0} + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi^*(x)} \frac{d\phi^{*'}(x)}{da} \Big|_{a=0} \quad (1.15)$$

$$J^\mu = -\partial^\mu \phi^* \frac{d\phi'(x)}{da} \Big|_{a=0} - \partial^\mu \phi \frac{d\phi^{*'}(x)}{da} \Big|_{a=0} \quad (1.16)$$

$$J^\mu = -\partial^\mu \phi^* (i\phi) - \partial^\mu \phi (-i\phi^*) \quad (1.17)$$

$$J^\mu = i\phi^* \partial^\mu \phi - i\phi \partial^\mu \phi^* \quad (1.18)$$

it also can be defined with an overall minus sign. The conserved charge is,

$$Q = \int d^3\mathbf{x} J^0 = \int d^3\mathbf{x} (i\phi^* \partial^0 \phi - i\phi \partial^0 \phi^*) \quad (1.19)$$

$$Q = - \int d^3\mathbf{x} (i\phi^* \pi^* - i\phi \pi) \quad (1.20)$$

$$Q = i \int d^3\mathbf{x} (\pi \phi - \pi^* \phi^*) \quad (1.21)$$

1.C)

The Heisenberg equation of motion is,

$$[A(t, \mathbf{x}), H(t)] = i\dot{A}(t, \mathbf{x}) \quad (1.22)$$

applying to ϕ and substituting the already found expression for H ,

$$i\dot{\phi}(t, \mathbf{x}) = \left[\phi(t, \mathbf{x}), \int d^3\mathbf{y} [\pi^*(t, \mathbf{y})\pi(t, \mathbf{y}) + \nabla\phi^*(t, \mathbf{y}) \cdot \nabla\phi(t, \mathbf{y}) + m^2\phi^*(t, \mathbf{y})\phi(t, \mathbf{y})] \right] \quad (1.23)$$

$$i\dot{\phi}(t, \mathbf{x}) = \int d^3\mathbf{y} [\phi(t, \mathbf{x}), \pi^*(t, \mathbf{y})\pi(t, \mathbf{y}) + \nabla\phi^*(t, \mathbf{y}) \cdot \nabla\phi(t, \mathbf{y}) + m^2\phi^*(t, \mathbf{y})\phi(t, \mathbf{y})] \quad (1.24)$$

Remember that,

$$[\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = [\phi(t, \mathbf{x}), \phi^*(t, \mathbf{y})] = 0 \Rightarrow [\phi(t, \mathbf{x}), \nabla\phi(t, \mathbf{y})] = [\phi(t, \mathbf{x}), \nabla\phi^*(t, \mathbf{y})] = 0 \quad (1.25)$$

then,

$$i\dot{\phi}(t, \mathbf{x}) = \int d^3\mathbf{y} [\phi(t, \mathbf{x}), \pi^*(t, \mathbf{y})\pi(t, \mathbf{y})] \quad (1.26)$$

$$i\dot{\phi}(t, \mathbf{x}) = \int d^3\mathbf{y} \pi^*(t, \mathbf{y})[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] + \int d^3\mathbf{y} [\phi(t, \mathbf{x}), \pi^*(t, \mathbf{y})]\pi(t, \mathbf{y}) \quad (1.27)$$

$$i\dot{\phi}(t, \mathbf{x}) = \int d^3\mathbf{y} \pi^*(t, \mathbf{y})[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] \quad (1.28)$$

$$i\dot{\phi}(t, \mathbf{x}) = \int d^3\mathbf{y} \pi^*(t, \mathbf{y}) i\delta^{(3)}(\mathbf{x} - \mathbf{y}) \quad (1.29)$$

$$i\dot{\phi}(t, \mathbf{x}) = i\pi^*(t, \mathbf{x}) \Rightarrow \dot{\phi}(t, \mathbf{x}) = \pi^*(t, \mathbf{x}) \quad (1.30)$$

Analogously is possible to show that,

$$\dot{\phi}^* = \pi, \quad \dot{\pi}^* = \nabla^2\phi - m^2\phi, \quad \dot{\pi} = \nabla^2\phi^* - m^2\phi^* \quad (1.31)$$

1.D)

The canonical commutation relations are,

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad [\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = [\pi(t, \mathbf{x}), \pi(t, \mathbf{y})] = 0 \quad (1.32)$$

$$[\phi^*(t, \mathbf{x}), \pi^*(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad [\phi^*(t, \mathbf{x}), \phi^*(t, \mathbf{y})] = [\pi^*(t, \mathbf{x}), \pi^*(t, \mathbf{y})] = 0 \quad (1.33)$$

$$[\phi^*(t, \mathbf{x}), \phi(t, \mathbf{y})] = [\pi^*(t, \mathbf{x}), \pi(t, \mathbf{y})] = 0 \quad (1.34)$$

We expand the field in it's inverse Fourier transform,

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) e^{ix \cdot p} \quad (1.35)$$

The equation of motion is,

$$(-\square_x + m^2)\phi(x) = 0 \quad (1.36)$$

which when acting on the momentum space representation gives,

$$0 = (-\square_x + m^2)\phi(x) = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) (-\square_x + m^2) e^{ix \cdot p} \quad (1.37)$$

$$0 = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) (p^2 + m^2) e^{ix \cdot p} \quad (1.38)$$

this integral equation should be valid for every x . In fact, given an arbitrary function $\psi(x)$, we can integrate over x ,

$$0 \int d^4 x \psi(x) = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) (p^2 + m^2) \int d^4 x e^{ix \cdot p} \psi(x) \quad (1.39)$$

$$0 = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) (p^2 + m^2) \tilde{\psi}(-p) \quad (1.40)$$

as ψ is arbitrary, the only way to the result be zero is if $\tilde{\phi}(p)(p^2 + m^2)$ is itself the zero distribution. Naively this would imply $\tilde{\phi}(p) \equiv 0$, but, due to the distributional identity $x\delta(x) = 0$, the most general solution is $\tilde{\phi}(p) \propto \delta(p^2 + m^2)$. The most general solution of this equation is $\tilde{\phi}(p) = 2\pi\delta(p^2 + m^2)(\theta(p^0)a(\mathbf{p}) + \theta(-p^0)b^*(\mathbf{p}))$. Thus,

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^4} \tilde{\phi}(p) e^{ix \cdot p} \quad (1.41)$$

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^4} 2\pi\delta(p^2 + m^2) (\theta(p^0)a(\mathbf{p}) + \theta(-p^0)b^*(\mathbf{p})) e^{ix \cdot p} \quad (1.42)$$

to simplify this we need to use some additional properties of the Dirac delta,

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^3 2\omega_{\mathbf{p}}} (\delta(p^0 - \omega_{\mathbf{p}}) + \delta(p^0 + \omega_{\mathbf{p}})) (\theta(p^0)a(\mathbf{p}) + \theta(-p^0)b^*(\mathbf{p})) e^{ix \cdot p} \quad (1.43)$$

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^3 2\omega_{\mathbf{p}}} (\delta(p^0 - \omega_{\mathbf{p}})a(\mathbf{p}) + \delta(p^0 + \omega_{\mathbf{p}})b^*(\mathbf{p})) e^{ix \cdot p} \quad (1.44)$$

$$\phi(x) = \int \frac{d^3 \mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} (a(\mathbf{p})e^{ix \cdot p} + b^*(\mathbf{p})e^{-ix \cdot p}) \quad (1.45)$$

Now we use the commutation relations of the fields to obtain the commutation relations of the creation and annihilation operators,

1.E)

Let's start with the Hamiltonian,

$$H = \int d^3 \mathbf{x} [\pi^* \pi + \nabla \phi^* \cdot \nabla \phi + m^2 \phi^* \phi] \quad (1.46)$$

$$\begin{aligned}
H = \int d^3\mathbf{x} \int \frac{d^3\mathbf{p} d^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} & \left[\partial_0 (a(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} + b^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) \partial_0 (b(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} + a^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \right. \\
& + \nabla (b(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} + a^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) \cdot \nabla (a(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} + b^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \\
& \left. + m^2 (b(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} + a^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) (a(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} + b^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \right] \quad (1.47)
\end{aligned}$$

$$\begin{aligned}
H = \int d^3\mathbf{x} \int \frac{d^3\mathbf{p} d^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} & \left[i\omega_{\mathbf{k}} (-a(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} + b^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) i\omega_{\mathbf{p}} (-b(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} + a^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \right. \\
& + i^2 \mathbf{p} \cdot \mathbf{k} (b(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} - a^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) (a(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} - b^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \\
& \left. + m^2 (b(\mathbf{k})e^{i\mathbf{x}\cdot\mathbf{k}} + a^\dagger(\mathbf{k})e^{-i\mathbf{x}\cdot\mathbf{k}}) (a(\mathbf{p})e^{i\mathbf{x}\cdot\mathbf{p}} + b^\dagger(\mathbf{p})e^{-i\mathbf{x}\cdot\mathbf{p}}) \right] \quad (1.48)
\end{aligned}$$

$$\begin{aligned}
H = \int d^3\mathbf{x} \int \frac{d^3\mathbf{p} d^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} & \left[e^{i\mathbf{x}\cdot(\mathbf{k}+\mathbf{p})} (-\omega_{\mathbf{k}}\omega_{\mathbf{p}} a(\mathbf{k})b(\mathbf{p}) - \mathbf{k} \cdot \mathbf{p} b(\mathbf{k})a(\mathbf{p}) + m^2 b(\mathbf{k})a(\mathbf{p})) \right. \\
& + e^{i\mathbf{x}\cdot(\mathbf{k}-\mathbf{p})} (\omega_{\mathbf{k}}\omega_{\mathbf{p}} a(\mathbf{k})a^\dagger(\mathbf{p}) + \mathbf{k} \cdot \mathbf{p} b(\mathbf{k})b^\dagger(\mathbf{p}) + m^2 b(\mathbf{k})b^\dagger(\mathbf{p})) \\
& + e^{i\mathbf{x}\cdot(-\mathbf{k}+\mathbf{p})} (\omega_{\mathbf{k}}\omega_{\mathbf{p}} b^\dagger(\mathbf{k})b(\mathbf{p}) + \mathbf{k} \cdot \mathbf{p} a^\dagger(\mathbf{k})a(\mathbf{p}) + m^2 a^\dagger(\mathbf{k})a(\mathbf{p})) \\
& \left. + e^{-i\mathbf{x}\cdot(\mathbf{k}+\mathbf{p})} (-\omega_{\mathbf{k}}\omega_{\mathbf{p}} b^\dagger(\mathbf{k})a^\dagger(\mathbf{p}) - \mathbf{k} \cdot \mathbf{p} a^\dagger(\mathbf{k})b^\dagger(\mathbf{p}) + m^2 a^\dagger(\mathbf{k})b^\dagger(\mathbf{p})) \right] \quad (1.49)
\end{aligned}$$

$$\begin{aligned}
H = \int \frac{d^3\mathbf{p}}{(2\pi)^3 4\omega_{\mathbf{p}}^2} & \left[e^{-i\mathbf{x}^0 2\omega_{\mathbf{p}}} (-\omega_{\mathbf{p}}\omega_{\mathbf{p}} a(-\mathbf{p})b(\mathbf{p}) + \mathbf{p}^2 b(-\mathbf{p})a(\mathbf{p}) + m^2 b(-\mathbf{p})a(\mathbf{p})) \right. \\
& + (\omega_{\mathbf{p}}\omega_{\mathbf{p}} a(\mathbf{p})a^\dagger(\mathbf{p}) + \mathbf{p}^2 b(\mathbf{p})b^\dagger(\mathbf{p}) + m^2 b(\mathbf{p})b^\dagger(\mathbf{p})) \\
& + (\omega_{\mathbf{p}}\omega_{\mathbf{p}} b^\dagger(\mathbf{p})b(\mathbf{p}) + \mathbf{p}^2 a^\dagger(\mathbf{p})a(\mathbf{p}) + m^2 a^\dagger(\mathbf{p})a(\mathbf{p})) \\
& \left. + e^{i\mathbf{x}^0 2\omega_{\mathbf{p}}} (-\omega_{\mathbf{p}}\omega_{\mathbf{p}} b^\dagger(-\mathbf{p})a^\dagger(\mathbf{p}) + \mathbf{p}^2 a^\dagger(-\mathbf{p})b^\dagger(\mathbf{p}) + m^2 a^\dagger(-\mathbf{p})b^\dagger(\mathbf{p})) \right] \quad (1.50)
\end{aligned}$$

$$H = \int \frac{d^3\mathbf{p}}{(2\pi)^3 4\omega_{\mathbf{p}}^2} \left[\omega_{\mathbf{p}}^2 a(\mathbf{p})a^\dagger(\mathbf{p}) + \omega_{\mathbf{p}}^2 b(\mathbf{p})b^\dagger(\mathbf{p}) + \omega_{\mathbf{p}}^2 b^\dagger(\mathbf{p})b(\mathbf{p}) + \omega_{\mathbf{p}}^2 a^\dagger(\mathbf{p})a(\mathbf{p}) \right] \quad (1.51)$$

$$H = \int \frac{d^3\mathbf{p}}{(2\pi)^3 4\omega_{\mathbf{p}}} \omega_{\mathbf{p}} \left[2(2\pi)^3 2\omega_{\mathbf{p}} \delta^{(3)}(0) + 2b^\dagger(\mathbf{p})b(\mathbf{p}) + 2a^\dagger(\mathbf{p})a(\mathbf{p}) \right] \quad (1.52)$$

$$H = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} \omega_{\mathbf{p}} \left[b^\dagger(\mathbf{p})b(\mathbf{p}) + a^\dagger(\mathbf{p})a(\mathbf{p}) \right] + 2\delta^{(3)}(0) \int d^3\mathbf{p} \frac{1}{2} \omega_{\mathbf{p}} \quad (1.53)$$

1.F)

$$\Delta_F(x, y) = \langle 0 | T \{ \phi(x) \phi^\dagger(y) \} | 0 \rangle \quad (1.54)$$

$$\Delta_F(x, y) = \theta(x^0 - y^0) \langle 0 | \phi(x) \phi^\dagger(y) | 0 \rangle + \theta(y^0 - x^0) \langle 0 | \phi^\dagger(y) \phi(x) | 0 \rangle \quad (1.55)$$

$$\Delta_F(x, y) = \int \frac{d^3\mathbf{p} d^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} \left(\theta(x^0 - y^0) \langle 0 | a(\mathbf{p}) e^{i\mathbf{x}\cdot\mathbf{p}} a^\dagger(\mathbf{k}) e^{-i\mathbf{y}\cdot\mathbf{k}} | 0 \rangle + \theta(y^0 - x^0) \langle 0 | b(\mathbf{k}) e^{i\mathbf{y}\cdot\mathbf{k}} b^\dagger(\mathbf{p}) e^{-i\mathbf{x}\cdot\mathbf{p}} | 0 \rangle \right) \quad (1.56)$$

$$\begin{aligned}
\Delta_F(x, y) = \int \frac{d^3\mathbf{p} d^3\mathbf{k}}{(2\pi)^6 4\omega_{\mathbf{p}}\omega_{\mathbf{k}}} & \left(\theta(x^0 - y^0) e^{-i\mathbf{y}\cdot\mathbf{k} + i\mathbf{x}\cdot\mathbf{p}} (2\pi)^3 2\omega_{\mathbf{p}} \delta^{(3)}(\mathbf{k} - \mathbf{p}) \langle 0 | 0 \rangle \right. \\
& \left. + \theta(y^0 - x^0) e^{-i\mathbf{x}\cdot\mathbf{p} + i\mathbf{y}\cdot\mathbf{k}} (2\pi)^3 2\omega_{\mathbf{p}} \delta^{(3)}(\mathbf{k} - \mathbf{p}) \langle 0 | 0 \rangle \right) \quad (1.57)
\end{aligned}$$

$$\Delta_F(x, y) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} \left(\theta(x^0 - y^0) e^{i(x-y)\cdot\mathbf{p}} + \theta(y^0 - x^0) e^{-i(x-y)\cdot\mathbf{p}} \right) \quad (1.58)$$

$$\Delta_F(x, y) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} e^{i(\mathbf{x}-\mathbf{y})\cdot\mathbf{p}} \left(\theta(x^0 - y^0) e^{-i(x^0-y^0)\omega_{\mathbf{p}}} + \theta(y^0 - x^0) e^{i(x^0-y^0)\omega_{\mathbf{p}}} \right) \quad (1.59)$$

Let us use the integral representation of the Heaviside function,

$$\theta(x) = -\frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} dp \frac{e^{-ixp}}{p + i\epsilon} \quad (1.60)$$

then,

$$\Delta_F(x, y) = -\frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^3 \mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} e^{i(\mathbf{x}-\mathbf{y}) \cdot \mathbf{p}} \int dp^0 \frac{1}{p^0 + i\epsilon} \left(e^{-i(x^0-y^0)p^0} e^{-i(x^0-y^0)\omega_{\mathbf{p}}} + e^{i(x^0-y^0)p^0} e^{i(x^0-y^0)\omega_{\mathbf{p}}} \right) \quad (1.61)$$

shifting the p^0 integral,

$$\Delta_F(x, y) = -\frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^3 \mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} e^{i(\mathbf{x}-\mathbf{y}) \cdot \mathbf{p}} \int dp^0 \frac{1}{p^0 - \omega_{\mathbf{p}} + i\epsilon} \left(e^{-i(x^0-y^0)p^0} + e^{i(x^0-y^0)p^0} \right) \quad (1.62)$$

changing the sign of p^0 in the second term,

$$\Delta_F(x, y) = -\frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^3 \mathbf{p}}{(2\pi)^3 2\omega_{\mathbf{p}}} e^{i(\mathbf{x}-\mathbf{y}) \cdot \mathbf{p}} \int dp^0 e^{-i(x^0-y^0)p^0} \left(\frac{1}{p^0 - \omega_{\mathbf{p}} + i\epsilon} + \frac{1}{-p^0 - \omega_{\mathbf{p}} + i\epsilon} \right) \quad (1.63)$$

$$\Delta_F(x, y) = -\frac{1}{i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4 2\omega_{\mathbf{p}}} e^{i(x-y) \cdot p} \left(\frac{-2\omega_{\mathbf{p}} + 2i\epsilon}{(p^0 - \omega_{\mathbf{p}} + i\epsilon)(-p^0 - \omega_{\mathbf{p}} + i\epsilon)} \right) \quad (1.64)$$

$$\Delta_F(x, y) = \frac{1}{i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{e^{i(x-y) \cdot p}}{-(p^0)^2 + \omega_{\mathbf{p}}^2 - 2i\epsilon\omega_{\mathbf{p}} - \epsilon^2} \quad (1.65)$$

$$\Delta_F(x, y) = \frac{1}{i} \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{e^{i(x-y) \cdot p}}{p^2 + m^2 - i\epsilon} \quad (1.66)$$

A Conventions

A.1 Metric

There are two choices of metric,

$$g_- = \text{diag}(- \quad + \quad + \quad +), \quad g_+ = \text{diag}(+ \quad - \quad - \quad -) \quad (\text{A.1})$$

I will always use g_- . This changes a lot of signs in a lot of places, as example,

$$\mathcal{L} = g_+^{\mu\nu} \frac{1}{2} \partial_\mu \phi^* \partial_\nu \phi - V[\phi, \phi^*] = -g_-^{\mu\nu} \frac{1}{2} \partial_\mu \phi^* \partial_\nu \phi - V[\phi, \phi^*] \quad (\text{A.2})$$

Physics should be invariant under conventions, so they don't matter, as long as you know which conventions you are using and are consistent.

A.2 Levi-Civita

The Levi-Civita is the (unique) totally antisymmetric four-pseudo-tensor, $\epsilon_{\alpha\beta\mu\nu}$, there are two possible choices of normalization,

$$\epsilon_{0123}^+ = 1, \quad \epsilon_{0123}^- = -1 \quad (\text{A.3})$$

I will always use ϵ^+ . We define also the spatial Levi-Civita by restricting the covariant one,

$$\epsilon_{ijk} := \epsilon_{0ijk} \quad (\text{A.4})$$

A.3 Creation and annihilation operators

When expanding the field in normal modes, there is a choice of normalization of the integral,

$$\phi(x) = \int \frac{d^3\mathbf{p}}{N_{\mathbf{p}}} (a(\mathbf{p})e^{ix \cdot p} + b^\dagger(\mathbf{p})e^{-ix \cdot p}) \quad (\text{A.5})$$

some common choices are,

$$N_{\mathbf{p}} = \begin{cases} (2\pi)^3 2\omega_{\mathbf{p}} \\ (2\pi)^3 \sqrt{2\omega_{\mathbf{p}}} \\ \sqrt{(2\pi)^3 2\omega_{\mathbf{p}}} \end{cases} \quad (\text{A.6})$$

this changes the commutation relation of the operators as,

$$[a(\mathbf{p}), a^\dagger(\mathbf{k})] = \frac{N_{\mathbf{p}}^2}{(2\pi)^3 2\omega_{\mathbf{p}}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) \quad (\text{A.7})$$

and also the definition of a momentum eigenstate,

$$|\mathbf{p}\rangle = \frac{(2\pi)^3 2\omega_{\mathbf{p}}}{N_{\mathbf{p}}} a^\dagger(\mathbf{p}) |0\rangle, \quad \langle \mathbf{k} | \mathbf{p} \rangle = 2\omega_{\mathbf{p}} (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}) \quad (\text{A.8})$$

I will always use $N_{\mathbf{p}} = (2\pi)^3 2\omega_{\mathbf{p}}$.